

MATH 4720 / MSSC 5720

Instructor: Mehdi Maadooliat

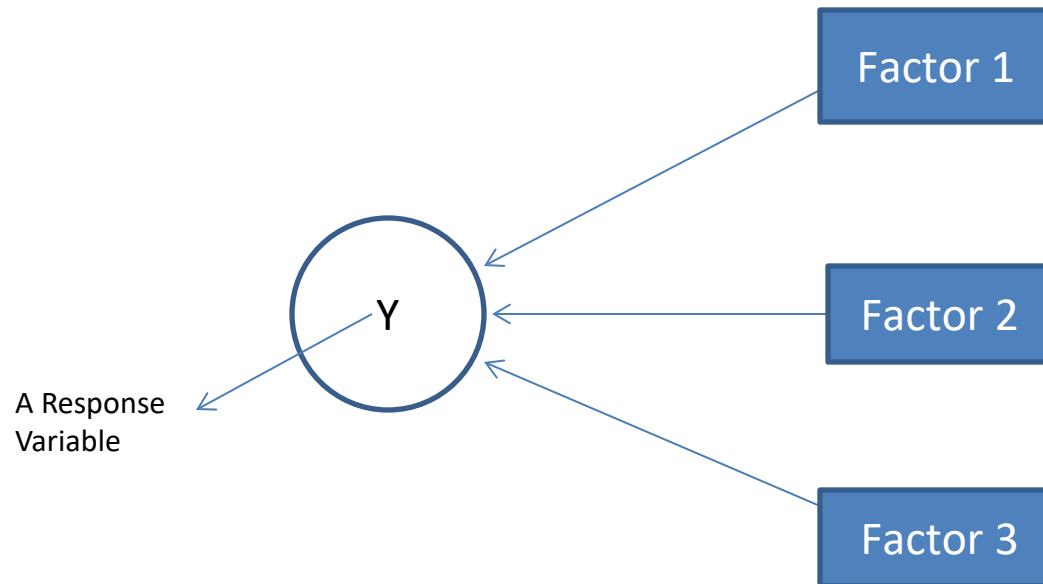
Chapter 8 (Part A)



Department of Mathematical and Statistical Sciences

ANALYSIS OF VARIANCE (ANOVA)

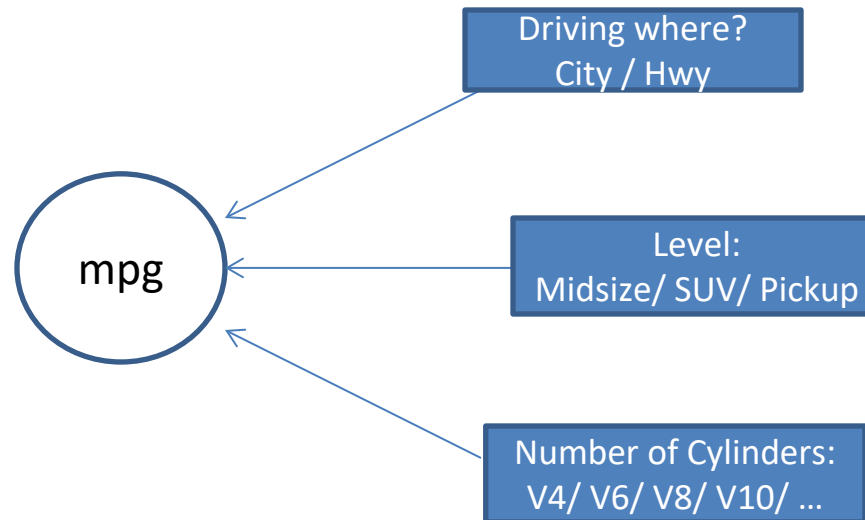
- **ANOVA is one of the most popular statistical tools of analyzing data.**



- **Does Y (the response) depends on any of the factors?**

ANOVA EXAMPLES

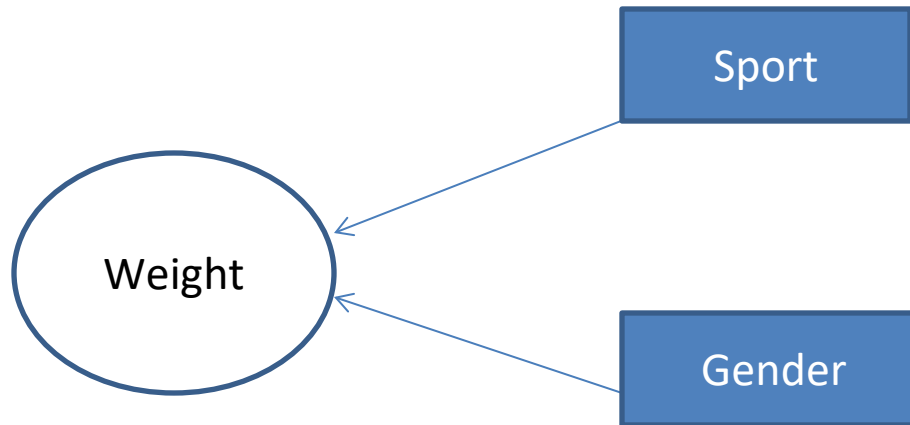
- **Example 1: You are doing a research on mpg (miles per gallon) for a brand of automobiles.**
- **Question: What effects mpg?**



- **Does Y (the response) depends on any of the factors?**

ANOVA EXAMPLES

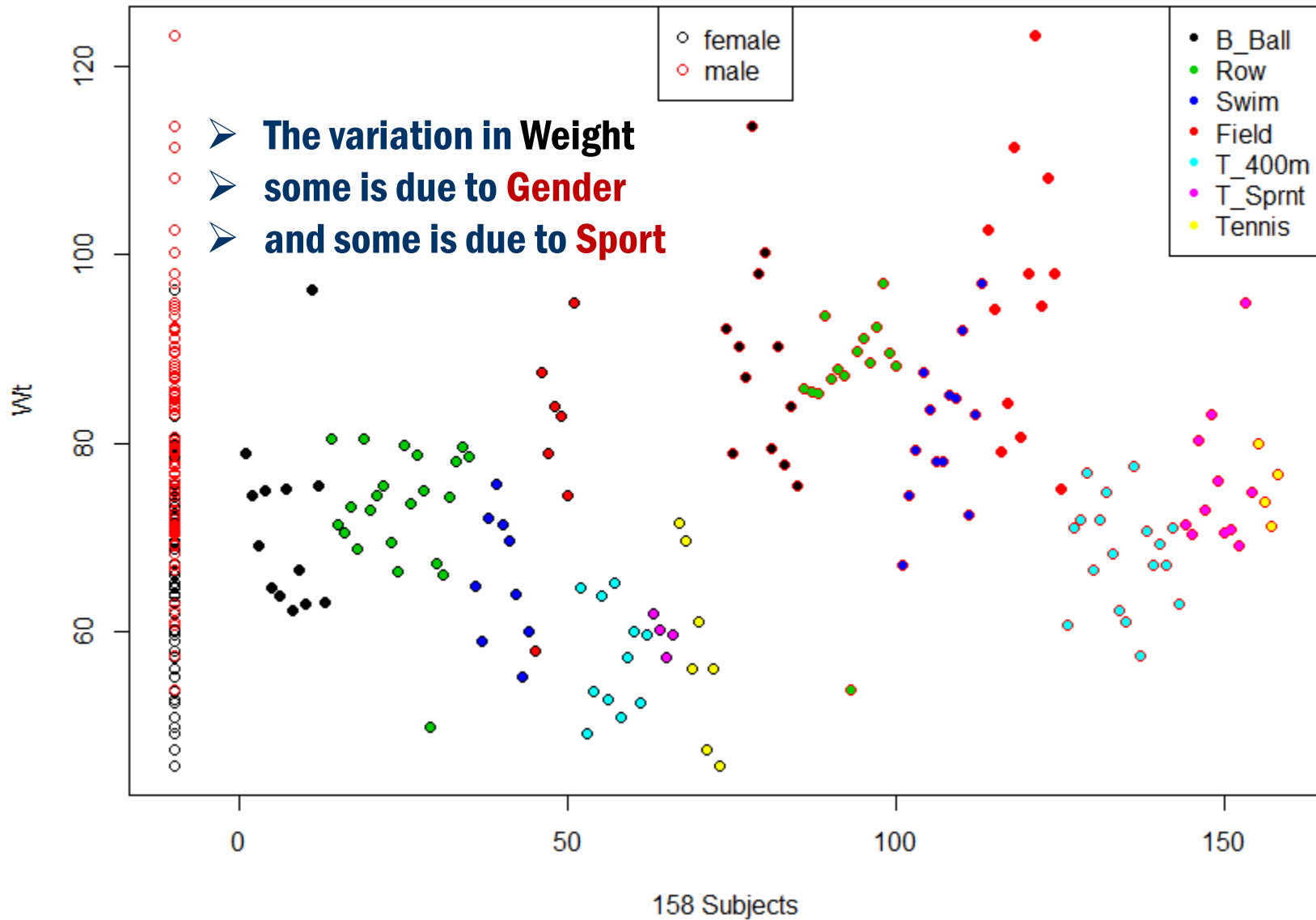
- **Example 2: (Australian Institute of Sport)**
- **Research Question: Does body weight (Wt) depend on**
 - Sport
 - Gender



- **In R (Australian Institute of Sport)**

EXAMPLE CONT'D

Australian Institute of Sport - Weight



CONCEPT

- **Variation(Wt) =**
Variation(Gender) + Variation(Sport)
+ Variation(Error)
- These variation can be described by **Sums of Squares** $\sum(\dots)^2$
- **$SS(Wt) = SS(Gender) + SS(Sport) + SS(Error)$**
- **$df_W = df_G + df_S + df_E$**
- **df is the degrees of freedom that represent the effective number of terms in the sums of squares**
- **In R:** `aov(Wt ~ sport + gender, data=ais2)`

TEST STATISTICS

- **F-Statistics**

- **Gender: Test Statistics**
$$F_1 = \frac{\frac{SS(\text{Gender})}{df_G}}{\frac{SS(\text{Error})}{df_E}} = \frac{MS_G}{MSE}$$

- If $F_1 > F_\alpha(df_G, df_E)$, then
gender is a significant factor

- **Sport: Test Statistic**
$$F_2 = \frac{\frac{SS(\text{Sport})}{df_S}}{\frac{SS(\text{Error})}{df_E}} = \frac{MS_S}{MSE}$$

- If $F_2 > F_\alpha(df_S, df_E)$, then
sport is a significant factor

BACK TO CONCEPT

- The sums of squares are not always easily available. For different factor-designs, there are different sums of squares.
- For One-Factor design, sums of squares are easy to compute.
- **One Factor ANOVA:**

	Treatment Levels				
	1	2	3	.	t
	y_{11}	y_{21}	y_{31}	.	y_{t1}
	y_{12}	y_{22}	y_{32}	.	y_{t2}
	.	.	.	.	.
	.	.	.	.	.
	y_{1n_1}	y_{2n_2}	y_{3n_3}	.	y_{tn_t}
	=====				
Mean	$\bar{y}_{1.}$	$\bar{y}_{2.}$	$\bar{y}_{3.}$	.	$\bar{y}_{t.}$
St.dev.	s_1	s_2	s_3	.	s_t



BACK TO CONCEPT

- Grand Mean:** $\bar{y}_{..} = \frac{n_1\bar{y}_{1.} + n_2\bar{y}_{2.} + \dots + n_t\bar{y}_{t.}}{n_1 + n_2 + \dots + n_t}$

- Total Variability:** $\sum_{i=1}^t \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{..})^2$

- Variability Between Samples:** $\sum_{i=1}^t n_i (\bar{y}_{i.} - \bar{y}_{..})^2$

- Variability Within Samples:** $\sum_{i=1}^t \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i.})^2$

- $\sum \sum (y_{ij} - \bar{y}_{..})^2 = \sum n_i (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum \sum (y_{ij} - \bar{y}_{i.})^2$
 $SS(Total) = SSB + SSE$

df's: $\sum n_i - 1$ $t - 1$ $\sum n_i - t$

$\nearrow$ $\nearrow$ $\nearrow$
 df_{Total} df_B df_E



BACK TO CONCEPT

$$\bullet \sum \sum (y_{ij} - \bar{y}_{..})^2 = \sum n_i (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum \sum (y_{ij} - \bar{y}_{i.})^2$$

$$SS(Total) = SSB + SSE$$

df's: $\sum n_i - 1$ $t - 1$ $\sum n_i - t$

$\nearrow$ $\nearrow$ $\nearrow$
 df_{Total} df_B df_E

- H_0 : There is no difference between the Treatments
- H_a : At least one of the treatment is different from the rest

- Test Statistics: $F = \frac{SS_B/df_B}{SS_E/df_E}$

- Decision Rule:

The Factor is significant if $F > F_\alpha(df_1 = df_B, df_2 = df_E)$

- [Applet](#)

FORMULATION IN TERMS OF HYPOTHESIS PROBLEM

- **One Factor ANOVA:**

Treatment Levels						
1	2	3	.	.	.	t
y_{11}	y_{21}	y_{31}	.	.	.	y_{t1}
y_{12}	y_{22}	y_{32}	.	.	.	y_{t2}
.	.	.	.	.	.	.
.	.	.	.	.	.	.
y_{1n_1}	y_{2n_2}	y_{3n_3}	.	.	.	y_{tn_t}
=====						
$N(\mu_1, \sigma_1^2)$	$N(\mu_2, \sigma_2^2)$	$N(\mu_3, \sigma_3^2)$	.	.	.	$N(\mu_t, \sigma_t^2)$

- $H_0: \mu_1 = \mu_2 = \cdots = \mu_t$
- $H_a: \mu_i \neq \mu_j$ **for some pairs** (i, j)



FORMULATION IN TERMS OF HYPOTHESIS PROBLEM CONT'D

- $H_0: \mu_1 = \mu_2 = \cdots = \mu_t$
- $H_a: \mu_i \neq \mu_j$ **for some pairs** (i, j)
- **Assumptions:**
 - $\sigma_1 = \sigma_2 = \cdots = \sigma_t$
 - **Data is generated from normal distribution for each treatment.**
- **TS** $F = \frac{SS_B/df_B}{SS_E/df_E}$
 - $SS_B = \sum n_i (\bar{y}_{i.} - \bar{y}_{..})^2$ $df_B = t - 1$
 - $SS_E = \sum (n_i - 1) s_i^2$ $df_E = \sum n_i - t$
- **Decision Rule:** **Reject** H_0 **in favor of** H_a **if**
 - $F > F_\alpha (df_B, df_E)$
- [Applet](#)

ANOVA TABLE

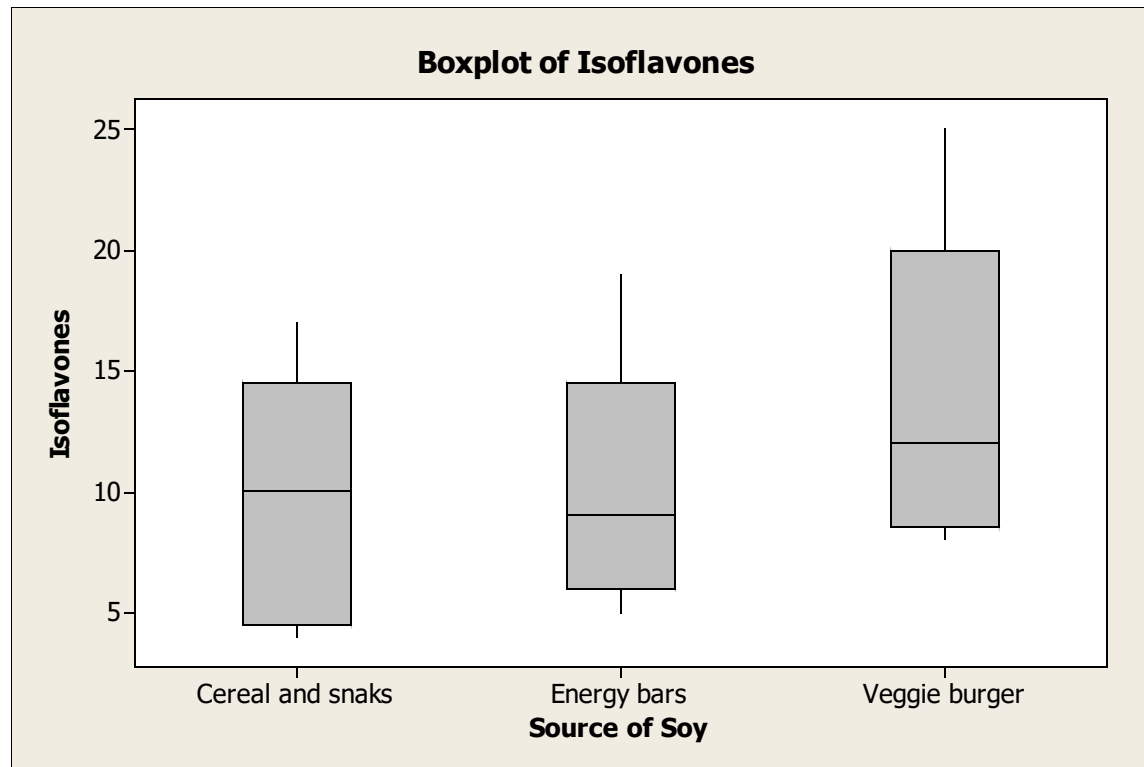
Source of Variation	df	Sum of Squares	Mean Square	F	p-value
Group (Between)	$t - 1$	$\sum n_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = SS_B$	$\frac{SS_B}{df_B} = MS_B$	$\frac{MS_B}{MS_E} = F_{\text{calc}}$	$\Pr(F > F_{\text{calc}})$
Error (Within)	$N - t$	$\sum (n_i - 1) s_i^2 = SS_E$	$\frac{SS_E}{df_E} = MS_E$		
Total	$N - 1$	$\sum (y_{ij} - \bar{y}_{\cdot\cdot})^2 = SS_T$			

- **Here:**

- $N = \sum_i^t n_i$
- $SS_T = SS_B + SS_E$
- MS_E is the pooled sample variance, an **estimate** for σ^2
- $R^2 = \frac{SS_B}{SS_T}$ is the proportion of the total variation explained by the **groups**.

BOOK EXAMPLE 8.1

- A hypothesis is that a nutrient “Isoflavones” varies among three types of food items: (1) Cereals and snacks, (2) energy bars, and (3) veggie burgers. A sample of five each is taken and the amount of isoflavones is measured.



EXAMPLE 8.1 CONT'D

- **Cereal and snacks:** $n_1 = 5$, $\bar{y}_1 = 9.20$, $s_1^2 = 33.7$
- **Energy bars:** $n_2 = 5$, $\bar{y}_2 = 10.00$, $s_2^2 = 29.0$
- **Veggie burger:** $n_3 = 5$, $\bar{y}_3 = 13.80$, $s_3^2 = 46.7$
- **Is there a sufficient evidence to conclude that the amount of isoflavones varies among these food items?** $\alpha = 0.05$.
- $H_0: \mu_1 = \mu_2 = \mu_3$
- $H_a: \mu_i \neq \mu_j$ **for some pairs** (i, j)
- **Assumptions:**
 - $\sigma_1 = \sigma_2 = \sigma_3$
 - **Data is generated from normal distribution for each type of food.**

EXAMPLE 8.1 ASSUMPTIONS

- $H_0: \sigma_1 = \sigma_2 = \sigma_3$
- In R: `car::leveneTest(exmp8.1$isof, exmp8.1$source)`
- `lawstat::levene.test(exmp8.1$isof, exmp8.1$source)`

Test for Equal Variances: Isoflavones versus Source of Soy

95% Bonferroni confidence intervals for standard deviations

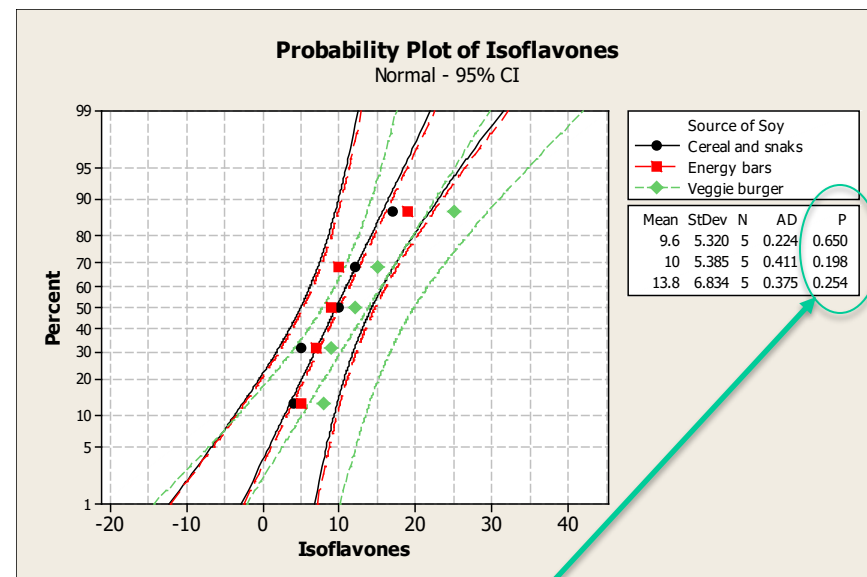
Source of Soy	N	Lower	StDev	Upper
Cereal and snaks	5	2.87498	5.31977	20.4752
Energy bars	5	2.91032	5.38516	20.7269
Veggie burger	5	3.69318	6.83374	26.3023

Bartlett's Test (Normal Distribution)
Test statistic = 0.30, p-value = 0.861

Levene's Test (Any Continuous Distribution)
Test statistic = 0.11, p-value = 0.896

- **Levene's test p – value is 0.896. Fail to reject equality of the variances**

- **H_0 : Data is generated from normal distribution for each type of food.**



- **Large p – values. Fail to reject Normality assumption.**

EXAMPLE 8.1 CONT'D

- $H_0: \mu_1 = \mu_2 = \mu_3$
- $H_a: \mu_i \neq \mu_j$ **for some pairs** (i, j)
- **TS** $F = \frac{SS_B/df_B}{SS_E/df_E}$
 - $SS_B = \sum n_i(\bar{y}_{i.} - \bar{y}_{..})^2$ $df_B = t - 1$
 - $SS_E = \sum (n_i - 1)s_i^2$ $df_E = \sum n_i - t$
- $\bar{y}_{..} = \frac{n_1\bar{y}_{1.} + n_2\bar{y}_{2.} + n_3\bar{y}_{3.}}{n_1 + n_2 + n_3} = \frac{5*9.2 + 5*10.0 + 5*13.8}{5+5+5} = 11.0$
- $df_B = (t - 1) = 3 - 1 = 2$
- $df_E = \sum n_i - t = (5 + 5 + 5) - 3 = 12$

EXAMPLE 8.1 CONT'D

- $SS_B = \sum n_i(\bar{y}_{i.} - \bar{y}_{..})^2$
 $= 5 * (9.2 - 11.0)^2 + 5 * (10.0 - 11.0)^2 + 5 * (13.8 - 11.0)^2$
 $= 60.40$
- $SS_E = \sum (n_i - 1)s_i^2$
 $= (5 - 1) * 33.0 + (5 - 1) * 29.0 + (5 - 1) * 46.7$
 $= 437.60$
- **TS** $F = \frac{SS_B/df_B}{SS_E/df_E} = \frac{60.40/2}{437.60/12} = 0.83$
- $F_\alpha(df_1 = 2, df_2 = 12) = 3.89$
- **Conclusion:** Is $F > 3.89$? No. Fail to reject H_0 . We cannot conclude that the amount of isoflavones vary among the food items.
- **F Calculator**

EXAMPLE 8.1 ANOVA TABLE

Source of Variation	df	Sum of Squares	Mean Square	F	p-value
Group (Between)	$t - 1$	$\sum n_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = SS_B$	$\frac{SS_B}{df_B} = MS_B$	$\frac{MS_B}{MS_E} = F_{\text{calc}}$	$\Pr(F > F_{\text{calc}})$
Error (Within)	$N - t$	$\sum (n_i - 1)s_i^2 = SS_E$	$\frac{SS_E}{df_E} = MS_E$		
Total	$N - 1$	$\sum (y_{ij} - \bar{y}_{\cdot\cdot})^2 = SS_T$			

- In R

➤ `summary(aov(isof ~ source, data=exmp8.1))`

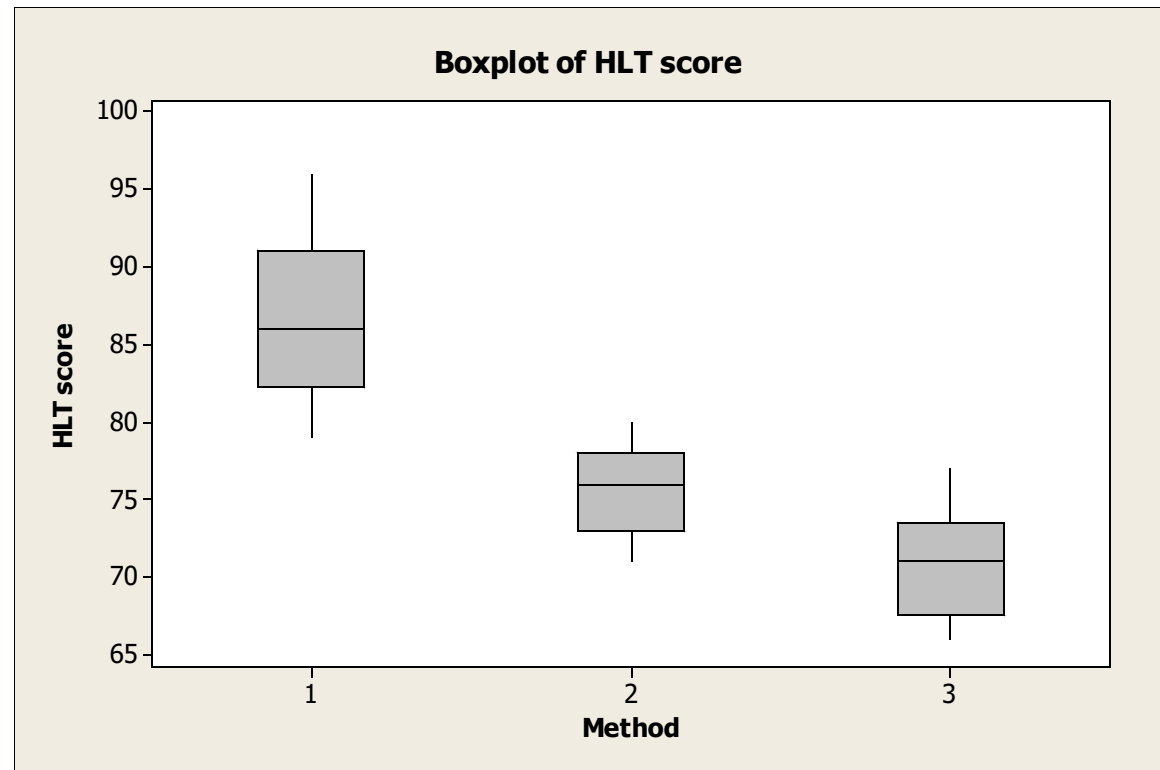
One-way ANOVA: Isoflavones versus Source of Soy

Source	DF	SS	MS	F	P
Source of Soy	2	60.4	30.2	0.83	0.460
Error	12	437.6	36.5		
Total	14	498.0			

S = 6.039 R-Sq = 12.13% R-Sq(adj) = 0.00%

BOOK EXAMPLE 8.2

- A clinical psychologist compares **three** methods for reducing the hospitality levels in university students. **HLT score** is used to measure the degree of hospitality. Randomly they assigned 8 students to method 1, 7 students to method 2 and 9 students to method 3. Each student was given HLT test at the end of semester



EXAMPLE 8.2 CONT'D

- **Is there a sufficient evidence to conclude difference among mean scores? Use $\alpha = 0.05$.**
- $H_0: \mu_1 = \mu_2 = \mu_3$
- $H_a: \mu_i \neq \mu_j$ **for some pairs (i, j)**
- **Assumptions:**
 - $\sigma_1 = \sigma_2 = \sigma_3$
 - **Data is generated from normal distribution for each type of food.**

EXAMPLE 8.2 ASSUMPTIONS

- $H_0: \sigma_1 = \sigma_2 = \sigma_3$
- In R: `car::leveneTest(exmp8.2$HLT, exmp8.2$method)`
- `lawstat::levene.test(exmp8.2$HLT, exmp8.2$method)`

Test for Equal Variances: HLT score versus Method

95% Bonferroni confidence intervals for standard deviations

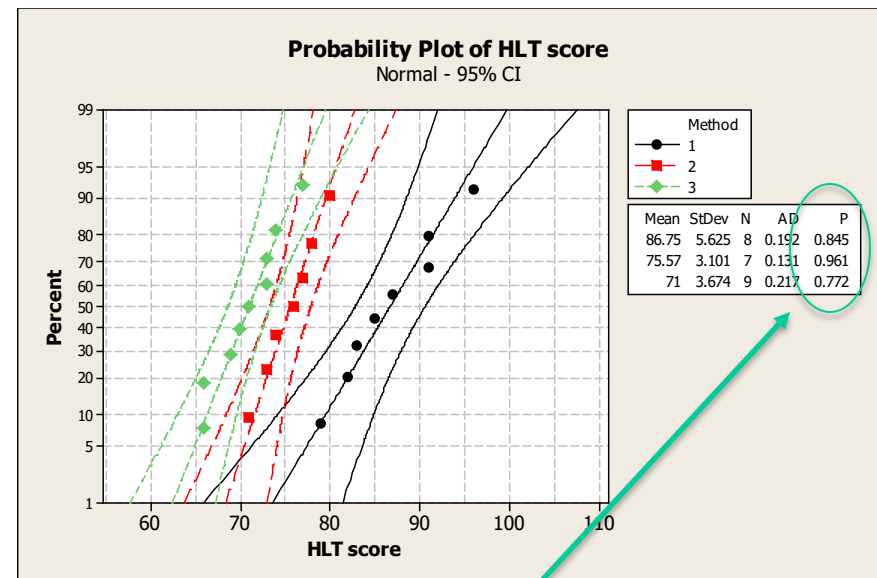
Method	N	Lower	StDev	Upper
1	8	3.41853	5.62520	13.7768
2	7	1.82797	3.10146	8.4154
3	9	2.29049	3.67423	8.3206

Bartlett's Test (Normal Distribution)
Test statistic = 2.46, p-value = 0.292

Levene's Test (Any Continuous Distribution)
Test statistic = 1.68, p-value = 0.210

- Levene's test p – value is 0.210. Fail to reject equality of the variances

- H_0 : Data is generated from normal distribution for each type of food.



- Large p – values. Fail to reject Normality assumption.

EXAMPLE 8.2 CONT'D

- $H_0: \mu_1 = \mu_2 = \mu_3$
- $H_a: \mu_i \neq \mu_j$ **for some pairs** (i, j)

One-way ANOVA: HLT score versus Method

Source	DF	SS	MS	F	P
Method	2	1090.6	545.3	29.57	0.000
Error	21	387.2	18.4		
Total	23	1477.8			

S = 4.294 R-Sq = 73.80% R-Sq(adj) = 71.30%

➤ **In R:** `summary(aov(HLT ~ method, data=exmp8.2))`

- **TS** $F = \frac{SS_B/df_B}{SS_E/df_E} = \frac{1090.6/2}{387.2/21} = 29.57$

- $F_\alpha(df_1 = 2, df_2 = 21) = 3.47$

- **Conclusion:** Is $F > 3.47$? **Yes. Reject H_0 .**

- [F Calculator](#)

NORMALITY ASSUMPTION FAILS

NON-PARAMETRIC METHOD

- In the ANOVA method, we assume that the sample from each treatment level is drawn from normal population. What if the distribution is non-normal.
- **The Kruskal-Wallis Test**
- H_0 : All t distributions are identical
- H_a : Not all distributions are the same.
- **TS:**
 1. Rank all samples from the lowest to the highest.
 2. The test statistics H is similar to the F-statistic based on the ranks.
- **Decision Rule:** Reject H_0 if $H > \chi^2_{\alpha}(df = t - 1)$
- **In R:**
 - `kruskal.test(x, g)`

WHAT IF EQUALITY OF THE VARIANCES FAIL?



- The assumption that the sample are generated from normal distribution is not very important as long as the total sample size is **large**.
- Note that conceptually the test statistic $F = \frac{SS_B/df_B}{SS_E/df_E}$ still makes sense.
- The major problem is with the assumption $\sigma_1 = \sigma_2 = \cdots = \sigma_t$. If this cannot be assumed, F- test **must not be used**.
- If $H_0: \sigma_1 = \sigma_2 = \cdots = \sigma_t$ is **rejected**, then one approach is to **transform** the data if the variances σ^2 is a function of the mean μ .